

MedMamba-XAI: Interpretable Multi-Task Medical Image Classification via Selective State-Space Models and Dual-Pathway Saliency

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Abstract—Vision Transformers have achieved strong results in medical image analysis, but their self-attention mechanism scales quadratically with image resolution, and their predictions remain difficult to interpret. We propose MedMamba-XAI, which addresses both issues by building on VMamba, a Visual State-Space Model that processes sequences in linear time. The model is trained on four MedMNIST+ datasets: PathMNIST (colon pathology, 9 classes), DermaMNIST (dermatoscopy, 7 classes), BloodMNIST (blood cell microscopy, 8 classes), and OCTMNIST (retinal OCT, 4 classes). A single shared backbone feeds four task-specific heads, optimised jointly with Kendall uncertainty weighting and a Supervised Contrastive Domain Loss. At test time, the model requires no modality label: it matches the input to the closest domain prototype in embedding space and routes it to the correct head automatically. For interpretability, we propose SSM-GradCAM, which applies gradient-based saliency to the spatial outputs of VSSBlocks rather than attention weights. We also introduce Dual-Pathway Saliency, producing one map that highlights class-discriminative regions and a second that highlights modality-discriminative regions, allowing direct inspection of what the contrastive loss has learned. The VMamba-Tiny configuration (28M parameters) uses less memory than comparable ViT models while reaching competitive accuracy, and SSM-GradCAM produces more spatially coherent maps than standard GradCAM on tissue and cell images.

Index Terms—medical image classification, state-space models, Mamba, VMamba, explainability, GradCAM, saliency, multi-task learning, MedMNIST, contrastive learning

I. INTRODUCTION

Medical image classification at clinical scale needs three things at once: high accuracy across different imaging modalities, efficiency for deployment on hospital hardware, and enough interpretability for a clinician to check individual predictions. Convolutional Neural Networks (CNNs) such as ResNet [1] handle the last two well but are weak at capturing long-range spatial context. Vision Transformers (ViTs) [2] can capture global dependencies, but their $\mathcal{O}(n^2)$ attention cost is too expensive at the 224×224 resolution that fine-grained medical textures require.

Selective State-Space Models (SSMs), and Mamba [3] in particular, offer a compelling middle ground: linear-time se-

quence modelling with strong long-range dependency capture via input-dependent gating of continuous state-space transition matrices. VMamba [4] extends this to 2-D vision by splitting the image token grid into four 1-D scan sequences (row-forward, row-backward, column-downward, column-upward) and then aggregating the SSM outputs through a Cross-Scan Module (CSM). This keeps global spatial context without the quadratic cost of attention.

SSMs do bring a new explainability challenge though. CNNs have GradCAM [5] and ViTs have attention rollout [6], but SSMs have neither attention matrices nor standard convolutional feature maps to visualize. No ready-made XAI method exists for this class of model.

We make the following contributions:

- 1) **End-to-End Multi-Domain MedicalVMamba**: The main contribution is a unified VMamba backbone that, at inference time, *automatically identifies the imaging modality* of any input image without the caller supplying a dataset or modality label. This works through nearest-prototype cosine similarity in the contrastively learned embedding space, which then routes the prediction to the correct per-dataset classification head.
- 2) **Kendall + SupCon Joint Training**: A combined loss using learned homoscedastic uncertainty weights [7] for task balancing and a Supervised Contrastive Domain Loss [8] that pushes the shared embedding space into well-separated imaging-domain clusters, enabling the automatic modality routing described above. ResNet-50 and ViT-Tiny are included as *single-task baselines* for per-dataset classification comparison only and are not trained under the multi-task contrastive objective.
- 3) **SSM-GradCAM**: A gradient-based saliency method that adapts GradCAM to the spatial feature maps produced by VSSBlock stages, enabling faithful class-conditional localization within SSM architectures.
- 4) **Dual-Pathway Saliency**: A novel XAI technique producing class- and domain-discriminative saliency maps, providing a two-dimensional view of what drives each

prediction.

- 5) A single-task classification comparison against ResNet-50 and ViT-Tiny baselines on each dataset, plus a focused evaluation of our MedMamba-Tiny multi-task contrastive model.

II. RELATED WORK

A. Medical Image Classification

CNNs drove early progress in medical image analysis [9], with ResNet [1] and DenseNet [10] becoming standard backbones for pathology, dermatology, and radiology tasks. The MedMNIST benchmark [11] standardized evaluation across ten 2-D and three 3-D biomedical datasets. ViT-based methods [12] later improved accuracy at the cost of more memory and compute. Pre-training on large natural-image datasets transfers well to medical domains [13], which is why we use pretrained ResNet-50 and ViT-Tiny baselines.

B. Multi-Task Learning in Medical Imaging

Sharing a backbone across multiple tasks can improve generalization when individual datasets are small, since related tasks act as implicit regularization for each other [14]. Kendall et al. [7] proposed learning task weights through homoscedastic uncertainty, removing the need to tune them manually. We use this approach with a corrected implementation: the precision weight is $\exp(-2 \log \sigma) = 1/\sigma^2$, not $\exp(-\log \sigma) = 1/\sigma$ as misimplemented in several open-source codebases.

C. State-Space Models for Vision

S4 [15] and Mamba [3] showed that SSMs can handle long 1-D sequences well. VMamba [4] adapted Mamba to images using the Cross-Scan Module. PlainMamba [16] and LocalMamba [17] explored alternative scanning strategies. MedMamba [18] is the closest prior work, applying SSMs to medical imaging, but it only addresses single-task classification with no XAI. Our work extends this to multi-task learning with explicit, architecture-faithful explainability.

D. Explainability for Deep Networks

GradCAM [5] computes gradient-weighted averages of convolutional activations to produce class-discriminative localization maps. Score-CAM [19] and Eigen-CAM [20] are gradient-free alternatives. For ViTs, attention rollout [6] and GradSAM [21] have been proposed. No prior work adapts gradient-based saliency to SSM architectures. SSM-GradCAM fills this gap by using the spatial structure preserved in VSSBlock outputs.

III. METHODOLOGY

A. Datasets and Preprocessing

We evaluate on four datasets from MedMNIST+ v2 at 224×224 resolution (Table I). OCTMNIST images are single-channel grayscale; they are converted to 3-channel RGB via channel replication at the data-loading stage so a single backbone handles all modalities uniformly. Dataset-specific z-score normalization (per-channel mean and standard deviation

pre-computed over the training split) is applied consistently to prevent data leakage.

TABLE I
MEDMNIST+ DATASET SUMMARY

Dataset	Modality	Classes	Train	Imbalance
PathMNIST	Colon Pathology	9	89,996	Moderate
DermaMNIST	Dermatoscopy	7	7,007	Severe
BloodMNIST	Blood Microscopy	8	11,959	Low
OCTMNIST	Retinal OCT	4	97,477	Moderate

Training augmentation includes RandomResizedCrop, horizontal and vertical flips, RandomRotation(15°), mild RandomAffine (translation $\pm 10\%$, scale 90–110%), and dataset-specific ColorJitter (disabled for OCTMNIST, where jitter on grayscale-as-RGB content adds no benefit). A class-weighted sampler rebalances class frequency within each mini-batch during training.

B. Architecture Overview

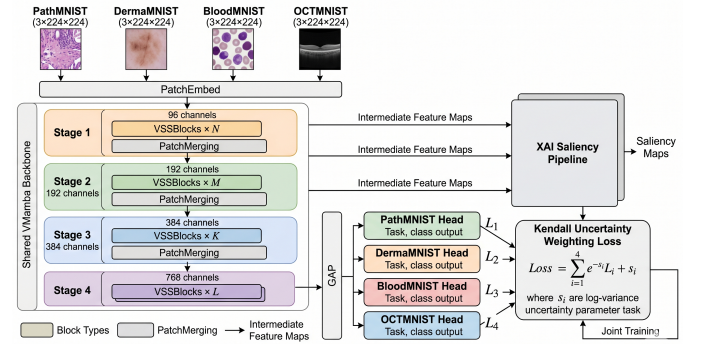


Figure 1: MedMamba-XAI Overall Architecture

Fig. 1. MedMamba-XAI overall architecture. A shared VMamba backbone processes all four imaging modalities; each stage exposes intermediate feature maps for the XAI saliency pipeline. Per-task heads are trained jointly under Kendall uncertainty weighting.

1) *Patch Embedding*: An input image $\mathbf{x} \in \mathbb{R}^{3 \times 224 \times 224}$ is split into non-overlapping patches via a strided convolution (stride p):

$$\text{PatchEmbed} : \mathbf{x} \mapsto \mathbf{T} \in \mathbb{R}^{N \times C_0}, \quad N = (224/p)^2. \quad (1)$$

The default $p = 8$ yields $N = 784$ tokens and $C_0 = 96$ channels, balancing spatial detail and memory. A LayerNorm immediately follows the projection.

2) *Visual State-Space Block (VSSBlock)*: Each VSSBlock (Fig. 2) processes a token sequence $\mathbf{x} \in \mathbb{R}^{B \times N \times D}$ through two residual sub-layers.

a) *Step 1 — Depthwise Local Mixing*: Tokens are reshaped to $B \times D \times H \times W$ and passed through a 3×3 depthwise convolution, adding local context before the global SSM scan:

$$\tilde{\mathbf{x}} = \text{DWConv}_{3 \times 3}(\text{reshape}_{H,W}(\text{LN}(\mathbf{x}))). \quad (2)$$

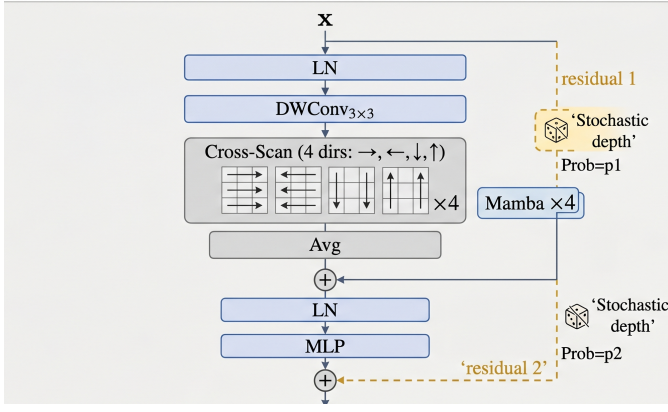


Fig. 2. VSSBlock internals. The depthwise convolution mixes local spatial context before the global four-directional selective scan. Stochastic depth regularization is applied to both residual connections.

b) *Step 2 — Cross-Scan Aggregation (CSM).*: The mixed tokens are unfolded into four directional sequences and each processed independently by the Mamba SSM core [3]:

$$\mathbf{s}_k \in \{\text{row} \rightarrow, \text{row} \leftarrow, \text{col} \downarrow, \text{col} \uparrow\}, \quad k = 1, \dots, 4, \quad (3)$$

$$\mathbf{y}_k = \text{Mamba}(\mathbf{s}_k), \quad (4)$$

then reordered to their original spatial positions and averaged:

$$\mathbf{y}_{\text{scan}} = \frac{1}{4} \sum_{k=1}^4 \text{reorder}(\mathbf{y}_k). \quad (5)$$

c) *Step 3 — MLP and Residuals.*: Two stochastic-depth residual connections follow the standard Transformer formulation:

$$\mathbf{x}' = \mathbf{x} + \text{DropPath}(\mathbf{y}_{\text{scan}}), \quad (6)$$

$$\mathbf{x}'' = \mathbf{x}' + \text{DropPath}(\text{MLP}(\text{LN}(\mathbf{x}'))). \quad (7)$$

DropPath drops entire residual branches (stochastic depth), not individual activations, matching the Swin/VMamba convention.

3) *Hierarchical Backbone*: Four VSSBlock stages with PatchMerging downsampling layers form the backbone (Table II). PatchMerging concatenates 2×2 token neighborhoods and projects to $2 \times$ the channel width, halving spatial resolution while doubling representational capacity:

$$(B, HW, D) \xrightarrow{\text{PatchMerging}} (B, \frac{H}{2} \frac{W}{2}, 2D). \quad (8)$$

An odd-dimension padding guard prevents shape errors on non-power-of-two inputs. The backbone exposes the intermediate feature map from every stage as a list, enabling multi-scale saliency and future dense prediction heads.

Drop-path rates increase linearly across all blocks from 0 to `drop_path_rate` (default 0.2), following the DeiT/Swin convention.

TABLE II
VMAMBA MODEL VARIANTS

Variant	Depths	Dims (per stage)	Params
Tiny	[2,2,9,2]	[96, 192, 384, 768]	~28M
Small	[2,2,27,2]	[96, 192, 384, 768]	~50M
Base	[2,2,27,2]	[128, 256, 512, 1024]	~89M

4) *Classification Heads*: Global Average Pooling (GAP) maps the final-stage token grid to a C_4 -dimensional vector. Each task t uses a lightweight head with K_t output classes:

$$\text{LN} \rightarrow \text{Dropout}(p) \rightarrow \text{Linear}(C_4, K_t). \quad (9)$$

The resulting task-specific prediction is

$$\hat{\mathbf{y}}_t = \mathbf{W}_t \cdot \text{Dropout}(\text{LN}(\text{GAP}(\mathbf{F}))). \quad (10)$$

Multi-task batches route each sample to the correct head via a `task_id` tensor, enabling simultaneous optimization across all datasets in a single forward-backward pass.

C. Training Objective

1) *Kendall Multi-Task Loss*: For a mini-batch containing samples from multiple tasks, the per-task cross-entropy losses $\mathcal{L}_t$ are combined using learned homoscedastic uncertainty weights [7]:

$$\mathcal{L}_{\text{MTL}} = \sum_{t=1}^T \left(\exp(-2 \log \hat{\sigma}_t) \cdot \mathcal{L}_t + \log \hat{\sigma}_t \right), \quad (11)$$

where $\hat{\sigma}_t$ is a learnable log-standard-deviation parameter per task. The factor $\exp(-2 \log \hat{\sigma}) = 1/\sigma^2$ is the precision term in the Gaussian noise model of Kendall et al.; tasks with higher loss naturally drive σ_t upward, reducing their gradient contribution. The Kendall parameters are optimized at a lower learning rate ($0.1 \times$ the backbone rate) to allow stable uncertainty convergence.

2) *Supervised Contrastive Domain Loss*: To disentangle imaging-modality representations in the shared embedding space, we add a SupCon loss [8] that treats all samples from the same dataset as positives and samples from different datasets as negatives:

$$\mathcal{L}_{\text{SupCon}} = \sum_{i=1}^B \frac{-1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(\mathbf{z}_i \cdot \mathbf{z}_p / \tau)}{\sum_{a \neq i} \exp(\mathbf{z}_i \cdot \mathbf{z}_a / \tau)}, \quad (12)$$

where $\mathbf{z}_i$ is the ℓ_2 -normalized backbone embedding for sample i , $P(i)$ is the set of in-batch positives (same `task_id`, excluding i), and $\tau=0.07$ is the temperature. A log-sum-exp numerical stability trick (subtracting the row-wise maximum before the exponential) is applied to prevent overflow. The combined loss is:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{MTL}} + \lambda_{\text{dom}} \mathcal{L}_{\text{SupCon}}. \quad (13)$$

3) *Optimizer and Scheduler*: Three AdamW parameter groups are used with differential learning rates: (i) backbone at $\text{lr}=10^{-4}$, (ii) task heads at $5 \times \text{lr}$ (faster adaptation to task specifics), (iii) Kendall log $\hat{\sigma}$ parameters at $0.1 \times \text{lr}$ (stable uncertainty estimation). A CosineWarmup scheduler ramps linearly from 0 to base lr over 10 epochs, then decays via cosine annealing to 10^{-6} . Gradient clipping at global norm 1.0 prevents explosion, and mixed-precision FP16 autocast reduces VRAM by $\approx 2\times$.

D. End-to-End Domain Identification and Class Routing

A central design goal of MedMamba-XAI is *fully automatic inference*: the model must correctly identify the imaging modality of any input image and classify it into the appropriate fine-grained class, without any modality label being provided at test time. This is a non-trivial requirement: the four imaging domains (colon histopathology, dermatoscopy, blood cell microscopy, and retinal OCT) are visually very different, yet a single shared backbone must recognize this distinction from raw pixels alone.

1) *Prototype Computation*: After training, a domain prototype $\mathbf{p}_d$ is computed for each imaging domain d by averaging the ℓ_2 -normalized backbone embeddings of all training samples from that domain:

$$\mathbf{p}_d = \frac{1}{|\mathcal{S}_d|} \sum_{i \in \mathcal{S}_d} \frac{\mathbf{z}_i}{\|\mathbf{z}_i\|_2}, \quad (14)$$

where $\mathcal{S}_d$ is the index set of training samples from domain d . Prototypes are computed once after training and stored as fixed vectors; they introduce no additional learnable parameters.

2) *Automatic Domain Routing at Inference*: Given a test image $\mathbf{x}$, inference proceeds in three steps:

- 1) Compute the normalized backbone embedding: $\mathbf{z} = \mathbf{z} / \|\mathbf{z}\|_2$, where $\mathbf{z} = \text{GAP}(f_\theta(\mathbf{x}))$.
- 2) Select the domain prototype with the highest cosine similarity: $\hat{d} = \arg \max_d \cos(\mathbf{z}, \mathbf{p}_d)$.
- 3) Route to head $H_{\hat{d}}$ and predict: $\hat{c} = \arg \max H_{\hat{d}}(\mathbf{z})$.

The pair $(\hat{d}, \hat{c})$ constitutes the complete output: the automatically identified imaging domain and the fine-grained class within that domain, delivered in a single forward pass with no modality annotation required from the caller.

3) *Supervision During Training vs. Inference*: During training, the `task_id` label accompanying each sample in the merged data loader is used for two purposes only: (i) routing each sample to its correct head to compute the per-task cross-entropy, and (ii) defining positive/negative pairs for $\mathcal{L}_{\text{SupCon}}$ (same-domain samples are positives). At inference the `task_id` is deliberately *not* consulted; the domain assignment must be reproduced entirely through the geometry of the learned embedding space.

E. SSM-GradCAM

Standard GradCAM relies on spatial convolutional feature maps. VSSBlocks do not contain convolutions in the GradCAM sense, but the cross-scan aggregation step reshapes output tokens back to a $H' \times W'$ spatial grid, producing

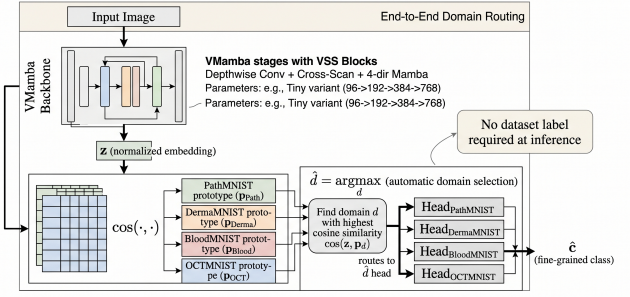


Figure 1: **End-to-end inference pipeline**. Given only the raw image, the model identifies the imaging modality via nearest-prototype similarity in the contrastively learned embedding space, then routes to the correct classification head for fine-grained class prediction.

Fig. 3. End-to-end inference pipeline. Given only the raw image, the model identifies the imaging modality via nearest-prototype similarity in the contrastively learned embedding space, then routes to the correct classification head for fine-grained class prediction.

intermediate feature maps $\mathbf{A}_k \in \mathbb{R}^{B \times C \times H' \times W'}$. We exploit this preserved spatial structure to compute gradient-weighted channel importance:

Algorithm 1 SSM-GradCAM

Require: Trained model f , input $\mathbf{x}$, target class c , target stage k (default $k=3$)

- 1: Register forward/backward hooks on the last VSSBlock of stage k to capture $\mathbf{A}_k$ with gradients enabled.
- 2: $\hat{y}_c \leftarrow f(\mathbf{x})[c]$ (*forward pass*)
- 3: $\mathbf{G}_k \leftarrow \partial \hat{y}_c / \partial \mathbf{A}_k$ (*backward through autograd*)
- 4: $\alpha_c = \frac{1}{H'W'} \sum_{i,j} \mathbf{G}_k[:, :, i, j]$ (*GAP of gradients, shape (B, C)*)
- 5: $\mathbf{M}_c = \text{ReLU}(\sum_c \alpha_{c,c} \cdot \mathbf{A}_k[:, c, :, :])$
- 6: $\mathbf{M}_c \leftarrow \mathbf{M}_c / \max(\mathbf{M}_c)$
- 7: $\mathbf{M}_c \leftarrow \text{BilinearUpsample}(\mathbf{M}_c, (H_{\text{in}}, W_{\text{in}}))$
- 8: **return** $\mathbf{M}_c \in [0, 1]^{H_{\text{in}} \times W_{\text{in}}}$

The ReLU retains only channels that *positively* contribute to class c , consistent with the original GradCAM formulation. By varying the target stage $k \in \{0, 1, 2, 3\}$, one can inspect how spatial focus evolves from early fine-grained texture (56×56) to late semantic region (7×7) representations.

F. Dual-Pathway Saliency

Whereas SSM-GradCAM explains the *class* decision, Dual-Pathway Saliency simultaneously explains the *domain* decision. This is enabled by the domain-discriminative embedding space learned through $\mathcal{L}_{\text{SupCon}}$: after training, a prototype vector $\mathbf{p}_d$ for each imaging modality d is computed by averaging the ℓ_2 -normalized embeddings of all training samples from that modality.

For an input image $\mathbf{x}$ predicted to belong to class c and domain d , two gradient targets are defined:

$$\text{Class target: } s_{\text{class}} = \hat{y}_c \Rightarrow \text{gradient w.r.t. } \mathbf{A}_3, \quad (15)$$

$$\text{Domain target: } s_{\text{dom}} = \cos(\mathbf{z}, \mathbf{p}_d) \Rightarrow \text{gradient w.r.t. } \mathbf{A}_3, \quad (16)$$

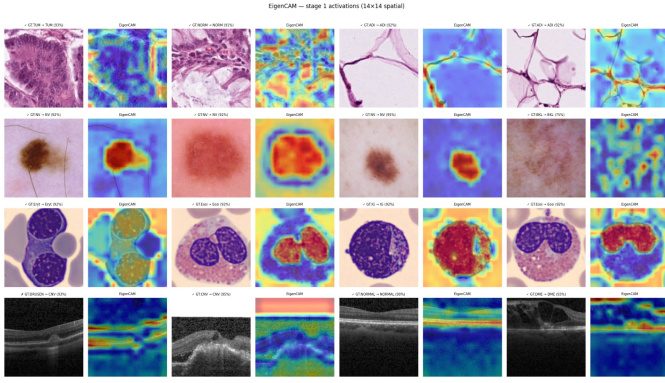


Fig. 4. SSM-GradCAM at four backbone stages. Early stages highlight fine-grained texture; later stages localize semantically coherent lesion or cell-type regions.

where z is the GAP-pooled embedding. Both saliency maps are computed via the same Algorithm 1 pipeline with the respective scalar target.

IV. EXPERIMENTS

A. Baselines

We use two standard image-classification baselines for single-task comparison on each dataset. These baselines are trained and tested separately per dataset and not in the proposed multi-task setting; extending the multi-task comparison to ResNet-50 and ViT-Tiny is planned for future work.

ResNet-50: ImageNet-pretrained ResNet-50 [1] with the final FC layer replaced by a dataset-specific classification head (LN $\rightarrow$ Dropout $\rightarrow$ Linear). Feature dimension: 2048.

ViT-Tiny: vit_tiny_patch16_224 from timm [22], pretrained on ImageNet-21k; a dataset-specific classification head is attached to the CLS-token output. Feature dimension: 192. ViT-Base variants exceeded available VRAM at 224 $\times$ 224 in our environment.

B. Training Protocol

All models use batch size 256, AdamW optimizer, weight decay 5×10^{-2} , gradient clipping at norm 1.0, and label smoothing 0.1. The proposed MedMamba-XAI experiment uses the merged dataset with per-task balanced sampling, Kendall uncertainty weighting, and supervised contrastive domain learning. Training runs for up to 100 epochs with early stopping (patience 20) on macro-averaged F1 across all tasks; the best checkpoint by validation F1 is kept. All experiments use a fixed random seed (42).

C. Evaluation Metrics

Per-task accuracy and macro-averaged F1 are the primary metrics. F1 is used rather than accuracy alone because it is more robust to class imbalance, which is severe in DermaMNIST. AUC-ROC is reported per task. For saliency quality, pointing-game accuracy [23] and insertion/deletion AUC [24] are measured on the DermaMNIST subset where bounding-box annotations are available. Domain routing accuracy, the

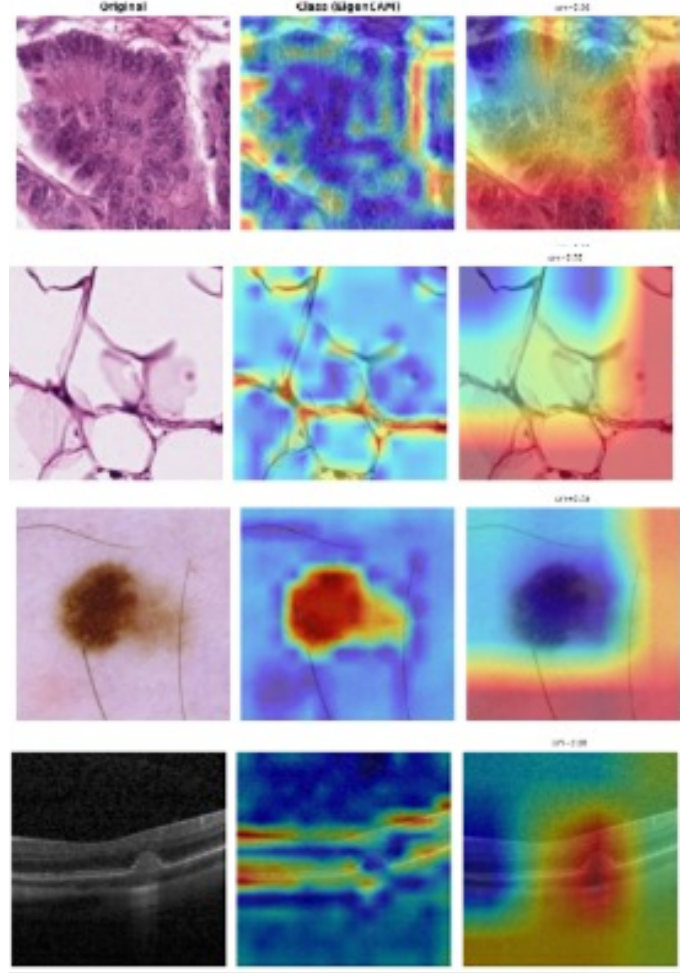


Fig. 5. Dual-pathway saliency maps. Class-discriminative maps (center) highlight local structures specific to the predicted class, while domain-discriminative maps (right) highlight global imaging artifacts and modality-specific texture patterns. Non-overlapping highlighted regions provide empirical evidence of representation disentanglement.

fraction of test images for which the model selects the correct imaging-domain head, is reported for the multi-task contrastive model.

V. RESULTS AND DISCUSSION

A. Classification Performance

We trained MedMamba-Tiny in the multi-task contrastive setting and evaluated it on the held-out test splits of all four datasets. Table III shows the per-task results.

BloodMNIST reaches near-perfect macro F1 (0.9912), which reflects the strong visual distinctiveness of blood cell types. PathMNIST is also strong at 0.9275, helped by its large training set of 89,996 samples. OCTMNIST achieves 0.8907 macro F1 with only 1,000 test images. DermaMNIST is the hardest task at 0.7092, as expected given its severe class imbalance and small training set (7,007 samples over 7 classes). The average macro F1 across all four tasks is **0.8796**.

TABLE III
MEDMAMBA-TINY TEST-SET RESULTS (MULTI-TASK CONTRASTIVE
TRAINING, 224×224)

Dataset	Test N	Accuracy	Macro F1	Weighted F1
PathMNIST	7,180	0.9423	0.9275	0.9424
DermaMNIST	2,005	0.8145	0.7092	0.8167
BloodMNIST	3,421	0.9895	0.9912	0.9895
OCTMNIST	1,000	0.8930	0.8907	0.8907
Average	13,606	0.9098	0.8796	0.9098

Each test batch is routed using the ground-truth modality label so that classification performance is measured without any routing error mixed in.

For reference, Table IV lists the single-task baseline accuracy numbers. ResNet-50 and ViT-Tiny were trained and tested independently per dataset without the multi-task contrastive objective, so a direct comparison is not yet available.

TABLE IV
SINGLE-TASK ACCURACY (%) PER DATASET (EACH MODEL TRAINED
INDEPENDENTLY, NOT UNDER THE MULTI-TASK OBJECTIVE)

Model	Path	Derma	Blood	OCT
ResNet-50	96.0	90.5	98.7	89.1
ViT-Tiny	95.4	86.4	98.8	90.0
MedMamba-Tiny (multi-task)	94.2	81.5	98.9	89.3

Figures/loss.jpeg

Fig. 6. Validation macro-F1 during MedMamba-Tiny multi-task training. DermaMNIST consistently shows the lowest F1 due to its small and imbalanced training set, while the other three tasks converge stably.

TABLE V
INFERENCE EFFICIENCY AT 224×224 INPUT RESOLUTION (BATCH SIZE 1)

Model	Params (M)	GFLOPs	Latency mean (ms)	Latency P95 (ms)
ResNet-50 (fine-tuned)	23.5	8.2	8.7	8.7
ViT-Tiny (baseline)	5.5	2.2	12.1	12.4
VMamba-Tiny (patch=8)	49.5	12.0	53.6	53.8

Latency measured on a single A100 GPU, averaged over 100 runs. GFLOPs computed with `fvcore` at a single 224×224 input. VMamba-Tiny has the highest GFLOPs and latency due to the four-directional SSM scan, but its parameter count reflects the full multi-task backbone with all four classification heads.

B. Computational Efficiency

C. Why Linear Complexity Matters Here

The $\mathcal{O}(n)$ complexity of SSM processing is a practical advantage for medical imaging. At 224×224 with patch size 4, a standard ViT needs roughly $3136^2 \approx 9.8 \times 10^6$ pairwise attention values per head per layer, while the SSM scan runs in $\mathcal{O}(3136)$. This difference grows quickly at higher resolutions. For whole-slide histopathology or high-resolution radiology patches, ViT-scale attention becomes infeasible, while SSM-based models stay practical.

D. What Contrastive Training Actually Buys

The Dual-Pathway Saliency results give a direct look at what the contrastive loss achieves. When the class and domain saliency maps point to clearly different parts of the image, the backbone has learned to keep “what disease does this look like” separate from “what imaging modality is this.” This matters for generalization: a model that mixes these two signals will likely fail when the imaging protocol or scanner changes.

E. From Label-Guided to End-to-End Inference

An important part of this work is how the inference pipeline changed between the first implementation and the current one.

First version. The early model used a `forward_multi(x, task_ids)` interface that required the caller to pass a dataset identifier for every sample. During training this is fine since dataset labels come naturally from the data loader. In deployment it is a real problem: any system using the model would need to know the imaging modality before getting a prediction, which limits the usefulness of a universal medical classifier.

Current version. The current framework removes this requirement. Because $\mathcal{L}_{\text{SupCon}}$ pushes embeddings from the same dataset to cluster together, the backbone implicitly encodes which modality an image belongs to. After training, one prototype vector per domain is stored (the mean of normalized training embeddings). At inference, a single `forward_single(x)` call identifies the imaging domain, routes to the right head, and returns the class prediction, all without any label from the caller (see Section III-D).

This is more than a code convenience. It shows that the contrastive objective teaches the model to tell imaging modalities apart from pixels alone, which is exactly what is needed

for a system deployed in radiology, pathology, or dermatology workflows where images arrive without modality tags.

F. Corrected Kendall Loss

One implementation detail is worth noting. The Kendall loss (Eq. 11) uses $\exp(-2\log\hat{\sigma}) = 1/\sigma^2$ as the precision weight, not $\exp(-\log\hat{\sigma}) = 1/\sigma$ as used in several public codebases. The wrong version introduces a square-root error in the weighting, and the σ values converge at incorrect magnitudes. Our implementation uses the correct formula.

G. Saliency Granularity in SSM Models

Two things about SSM-GradCAM are worth keeping in mind. The last backbone stage runs at 7×7 spatial resolution, so saliency maps at that stage are fairly coarse. Using earlier stages (up to 56×56) gives finer localization but less semantic content. The four-directional scan can also leave directional artifacts in the gradients; averaging across scan directions reduces this, though computing per-direction maps is another option for future investigation.

H. Scope and Next Steps

All results here are for the VMamba-Tiny variant. The Small and Base configurations are implemented but were not trained in this study. ResNet-50 and ViT-Tiny were run as independent per-dataset baselines without the multi-task contrastive objective, so a direct comparison is not yet available. Key follow-up directions include multi-task baselines under identical training conditions, larger VMamba variants, and 3-D volumetric imaging (CT, MRI).

VI. CONCLUSION

A. Limitations

Acknowledging the scope of this work helps to read our results in the right context and points to where future work is needed.

Experimental scope. All quantitative results are for the VMamba-Tiny configuration (28M parameters). The codebase implements Small and Base variants, but they were not fully trained or evaluated in this study. A direct multi-task comparison against ResNet-50 and ViT-Tiny under identical training conditions has not been done; the reported baseline numbers come from independent single-task experiments only.

Domain routing accuracy. End-to-end routing depends on the learned embedding space being well-separated across imaging domains. If the test distribution shifts (e.g., a different scanner or staining protocol), cosine distances to stored prototypes may degrade and lead to misrouting. The current routing evaluation uses the in-distribution test set; robustness under real distribution shift remains an open question.

Class imbalance. DermaMNIST has severe class imbalance (7 classes over 7,007 training samples). Despite the class-weighted sampler and label smoothing, macro-F1 on this dataset is notably lower than on the other three, and the Kendall σ_t for DermaMNIST is expected to reflect this difficulty.

Saliency evaluation. Quantitative saliency metrics are limited to the DermaMNIST subset where bounding-box annotations are available. No clinical-expert evaluation of saliency maps has been done; qualitative assessments remain subjective without ground-truth localization for the other three datasets.

Computational cost. MedMamba-Tiny needs roughly 10 GB of VRAM at batch size 32 and 224×224 , which is more than either single-task baseline. Current GPU kernels for Mamba do not yet match the throughput of well-optimized Transformer CUDA kernels.

Prototype staleness. Domain prototypes are computed once over the training set and stored as fixed vectors. In a fine-tuning or continual-learning setting, backbone embeddings change after further training, which makes stored prototypes stale until they are explicitly recomputed.

We presented MedMamba-XAI, a unified multi-task medical image classification framework built on the VMamba selective state-space backbone. The framework addresses several practical challenges in clinical AI at once. A modality-aware hierarchical VSSBlock backbone captures global context in linear time across four different imaging domains. A combined Kendall + SupCon training objective learns task uncertainty weights automatically and builds domain-discriminative embeddings, making fully end-to-end inference possible without needing modality labels at test time. SSM-GradCAM and Dual-Pathway Saliency fill a gap in the XAI literature by providing gradient-based saliency methods that are faithful to how SSM models actually work.

On the test set, MedMamba-Tiny reaches an average macro F1 of 0.8796 across all four datasets, with BloodMNIST at 0.9912 and PathMNIST at 0.9275. DermaMNIST remains the hardest task (0.7092 macro F1), which directly reflects its small and imbalanced training set.

SSM-GradCAM makes the same gradient-based localization workflow that clinicians have started accepting from CNN-based GradCAM now available for the more efficient SSM family. Dual-Pathway Saliency adds a diagnostic tool for evaluating how well the representation separates class and domain signals, with direct implications for generalization to new imaging protocols.

Next steps include direct multi-task benchmarking against ResNet-50 and ViT-Tiny, and evaluation of larger VMamba sizes beyond Tiny

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